**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
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# Math 102 Major Exam I Term 223

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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**1.** $\int_1^9 \frac{1}{\sqrt{x}\,(1+\sqrt{x})^2}\,dx =$

(a) $\frac{1}{9}$

(b) $3$

(c) $\frac{1}{2}$

(d) $\frac{1}{3}$

(e) $1$

**2.** The area of the region enclosed by $y = \frac{2}{\sqrt{4-x^2}}$, $y = \frac{-2}{\sqrt{4-x^2}}$, $x = -1$ and $x = 1$ equals to

(a) $\frac{2\pi}{3}$

(b) $\frac{4\pi}{3}$

(c) $4\pi$

(d) $\frac{4}{5}$

(e) $\frac{4\sqrt{3}}{2}$

**3.** Let $R$ be the region bounded by the parabola $y = 2x - x^2$ and the $x$-axis. If the line $y = mx$ divides the region $R$ into two regions of equal area, then $m =$

(a) $2 - \sqrt[3]{4}$

(b) $2 - \sqrt{3}$

(c) $1 - \sqrt[3]{2}$

(d) $1 - \sqrt{3}$

(e) $4 - \sqrt[3]{2}$

**4.** Let $f$ be an odd continuous function. If $\int_0^1 f(x)\,dx = 2$ and $\int_{-1}^3 f(x)\,dx = 4$, then $\int_{-3}^1 f(x)\,dx =$

(a) $-2$

(b) $2$

(c) $-4$

(d) $0$

(e) $4$

**5.** $\int_{-2}^3 \frac{2\,dx}{x^2+4x+8} =$

(a) $\ln\left(\frac{3}{2}\right)$

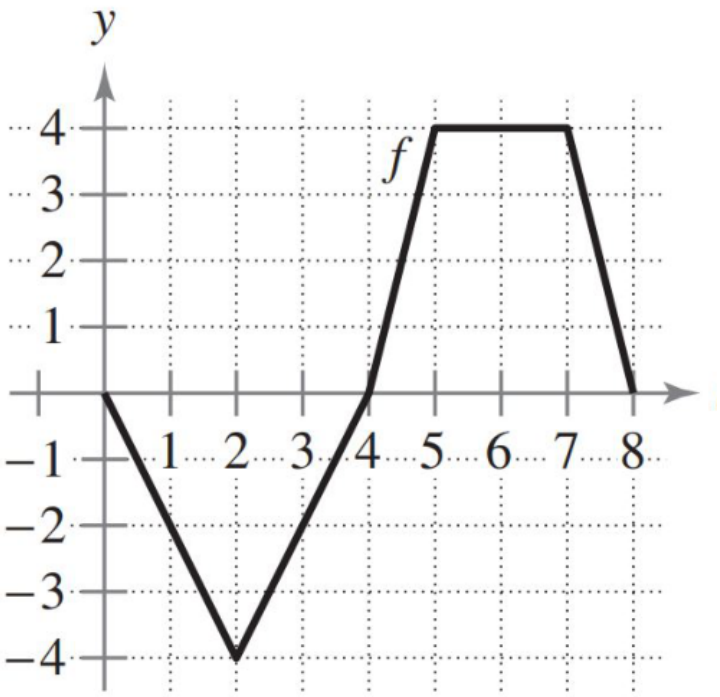
(b) $\tan^{-1}\left(\frac{5}{2}\right)$

(c) $2\cot^{-1}\left(\frac{3}{2}\right)$

(d) $\tan^{-1}\left(\frac{1}{2}\right)$

(e) $\tan^{-1}\left(\frac{3}{2}\right)$

**6.** Let $g(x) = \int_0^x f(t)\,dt$, where $f$ is the function whose graph is shown in the figure. The largest open interval, on which $g$ is decreasing, is




(a) $(0, 4)$

(b) $(0, 2) \cup (7, 8)$

(c) $(2, 5)$

(d) $(5, 7)$

(e) $(4, 5) \cup (7, 8)$

**7.** $\lim_{\|\Delta\|\to 0} \sum_{i=1}^{n} \sqrt{4 - c_i^2}\,\Delta x_i$ on $[0, 2]$ equals to

(a) $\frac{\pi}{4}$

(b) $\pi$

(c) $4\pi$

(d) $\frac{\pi}{2}$

(e) $2\pi$

**8.** The value(s) of $c$ guaranteed by the Mean Value Theorem for integrals for $f(x) = 5 - \dfrac{1}{x}$ over $[1, 4]$ is (are)

(a) $\dfrac{1}{\ln 4}$

(b) $\dfrac{4}{\ln 3}$ and $\dfrac{2}{\ln 3}$

(c) 1 and 4

(d) $\dfrac{3}{\ln 4}$

(e) $\ln\left(\dfrac{3}{2}\right)$

**9.** The upper sum, for the region bounded by the graph of $f(x) = 6 - 2x$ on $[1, 2]$ in terms of the number of sub-intervals $n$, is

(a) $3 - \dfrac{1}{n}$

(b) $3 + \dfrac{1}{n}$

(c) $3 + \dfrac{3}{n}$

(d) $3 - \dfrac{3}{n}$

(e) $3 + \dfrac{2}{3n}$

**10.** The area of the region bounded by the graphs of $f(x) = 2\sin x$ and $g(x) = \tan x$ over the interval $\left[-\dfrac{\pi}{3}, \dfrac{\pi}{3}\right]$ is

(a) 2

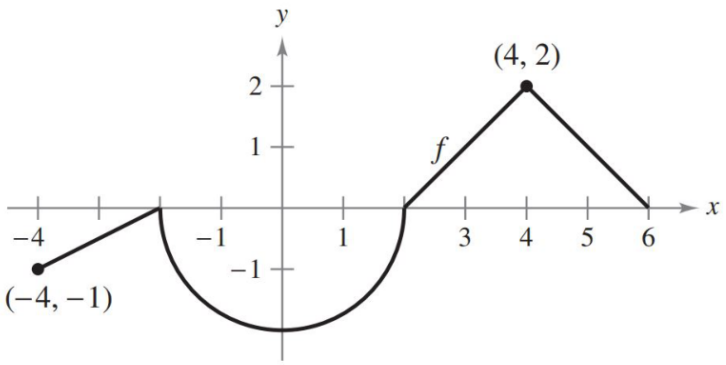
(b) $\dfrac{1}{2}$

(c) $2(1 - \ln 2)$

(d) $\dfrac{\ln 2}{2}$

(e) $2\ln 2 - 1$

**11.** If the graph of $f$ is shown in the figure, then $\int_{-4}^{6} \left(f(x) + \frac{\pi}{5}\right) dx =$




(a) 1

(b) $-4$

(c) 2

(d) 3

(e) 0

**12.** $\sum_{i=1}^{n} \left(\frac{2i^3-3i}{n^4}\right) - \frac{1}{n} + \frac{1}{n^2} + \frac{3}{2n^3} =$

(a) $\frac{1}{3}$

(b) 1

(c) $\frac{1}{4}$

(d) $\frac{1}{2}$

(e) $\frac{1}{8}$

**13.** If the average value of $f(x) = \dfrac{2\ln x}{x}$ over $[1, b]$ is $\dfrac{1}{e-1}$ then $b =$

(a) $1 + e$

(b) $e$

(c) $2e$

(d) $\frac{1}{e}$

(e) $\frac{2}{e}$

**14.** The area of the enclosed region between $x = 4 - y^2$ and $x = y - 2$ is

(a) $\dfrac{149}{5}$

(b) $\dfrac{25}{3}$

(c) $\dfrac{27}{6}$

(d) $\dfrac{125}{6}$

(e) $\dfrac{65}{6}$

**15.** $\int \frac{\csc^2 x}{\cot^3 x}\, dx =$

(a) $2\csc^2 x + C$

(b) $2\sin^2 x + C$

(c) $\dfrac{1}{2}\sec^2 x + C$

(d) $\tan^2 x + C$

(e) $\cot^2 x + C$

**16.** If $\int_1^b \frac{2}{t}\, dt = \int_{1/e}^b \frac{1}{t}\, dt$, then $b =$

(a) $\dfrac{1}{e}$

(b) $e$

(c) $2$

(d) $2e$

(e) $\dfrac{1}{2}$

**17.** An approximation, for the area between the graph of $f(x) = |\sin x|$, the $x$-axis, $x = -3\pi/4$ and $x = \pi/2$ using the left endpoints and 5 rectangles (of equal widths), is

(a) $\left(\dfrac{3\pi}{2}\right)\left(3\sqrt{2} - 1\right)$

(b) $\left(\dfrac{\pi}{4}\right)\left(3\sqrt{2} + 2\right)$

(c) $\left(\dfrac{\pi}{8}\right)\left(3\sqrt{2} + 2\right)$

(d) $\left(3\sqrt{2} + \pi\right)$

(e) $\left(\dfrac{\pi}{4}\right)\left(2\sqrt{2} + 1\right)$

**18.** $\int_0^{\ln 2} 2e^{-x} \sinh x \, dx =$

(a) $\ln 2 - \dfrac{3}{8}$

(b) $\dfrac{3}{8} - \ln 3$

(c) $e^2 - \ln 2$

(d) $\ln 2 + \dfrac{3}{8}$

(e) $\ln 2 - 1$

# Answer Key

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| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(c)** | $u = 1 + \sqrt{x}$: $\int_2^4 2u^{-2}du = -\frac{1}{2} + 1 = \frac{1}{2}$. |
| 2 | **(b)** | $A = \int_{-1}^{1} \frac{4}{\sqrt{4-x^2}}dx = 8 \arcsin \frac{1}{2} = \frac{4\pi}{3}$. |
| 3 | **(a)** | Area of $R = \frac{4}{3}$; $\frac{(2-m)^3}{6} = \frac{2}{3}$, so $m = 2 - \sqrt[3]{4}$. |
| 4 | **(c)** | Odd: $\int_{-1}^{1} f = 0$, so $\int_1^3 f = 4$; $\int_{-3}^{1} f = -\int_1^3 f = -4$. |
| 5 | **(b)** | Complete the square: $\left[\arctan \frac{x+2}{2}\right]_{-2}^{3} = \tan^{-1} \frac{5}{2}$. |
| 6 | **(a)** | $g' = f < 0$ exactly on $(0, 4)$, so $g$ decreases on $(0, 4)$. |
| 7 | **(b)** | Limit $= \int_0^2 \sqrt{4 - x^2}dx =$ quarter disc of radius $2 = \pi$. |
| 8 | **(d)** | Average $= 5 - \frac{\ln 4}{3}$; $5 - \frac{1}{c}$ equals it, so $c = \frac{3}{\ln 4}$. |
| 9 | **(b)** | $f$ decreasing, left endpoints: $S = \frac{1}{n}\sum(4 - \frac{2(i-1)}{n}) = 3 + \frac{1}{n}$. |
| 10 | **(c)** | Symmetry: $A = 2\int_0^{\pi/3}(2\sin x - \tan x)dx = 2(1 - \ln 2)$. |
| 11 | **(d)** | $\int f = -1 - 2\pi + 4 = 3 - 2\pi$ and $\int \frac{\pi}{5} = 2\pi$, total 3. |
| 12 | **(d)** | Sum formulas give $\frac{1}{2} + \frac{1}{n} - \frac{1}{n^2} - \frac{3}{2n^3}$; adding the rest leaves $\frac{1}{2}$. |
| 13 | **(b)** | $\int_1^b \frac{2\ln x}{x}dx = (\ln b)^2$; $\frac{(\ln b)^2}{b-1} = \frac{1}{e-1}$ holds for $b = e$. |
| 14 | **(d)** | Intersect at $y = -3, 2$: $A = \int_{-3}^{2}(6 - y - y^2)dy = \frac{125}{6}$. |
| 15 | **(c)** | $u = \cot x$: $-\int u^{-3}du = \frac{1}{2}\tan^2 x + C = \frac{1}{2}\sec^2 x + C'$. |
| 16 | **(b)** | $2\ln b = \ln b + 1 \Rightarrow \ln b = 1 \Rightarrow b = e$. |
| 17 | **(c)** | $\Delta x = \frac{\pi}{4}$, heights $\frac{\sqrt{2}}{2}, 1, \frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}$: $\frac{\pi}{8}(3\sqrt{2} + 2)$. |
| 18 | **(a)** | $2e^{-x}\sinh x = 1 - e^{-2x}$, antiderivative $x + \frac{1}{2}e^{-2x}$: result $\ln 2 - \frac{3}{8}$. |